

AI-BFSI Internship Project Documentation

AI Tools Used – Subscription Waste Detector

Introduction

Artificial Intelligence (AI) has transformed modern software development by assisting developers in designing, implementing, testing, and documenting systems. In the **Subscription Waste Detector**, AI was used in two distinct capacities:

- **ChatGPT** as a development assistant during planning, coding, debugging, documentation, and technical guidance.
- **Groq API** as an integrated recommendation engine within the application, generating personalized subscription cost-saving suggestions for end users.

This combination accelerated development, improved documentation quality, and enhanced application functionality, while ensuring all decisions and validations remained under developer supervision.

Objectives of Using AI Tools

- Accelerate development activities
- Improve code quality and efficiency
- Assist in debugging and troubleshooting
- Support architecture and design decisions
- Generate AI-powered financial recommendations
- Improve documentation quality
- Reduce repetitive tasks
- Enhance user experience

AI Tools Used in the Project

AI Tool	Purpose	Role in Project
ChatGPT	Development Assistance	Planning, coding, debugging, documentation, technical guidance
Groq API	AI Recommendation Engine	Generates personalized subscription cost-saving recommendations

ChatGPT as a Development Assistant

Overview: ChatGPT acted as an intelligent assistant throughout the project lifecycle. Its outputs were always reviewed, tested, and validated before adoption.

Areas of Usage:

- **Planning:** Brainstorming ideas, refining objectives, defining scope, structuring roadmap.
- **Development:** Streamlit components, Python logic, expense processing, subscription detection, dashboard improvements.
- **Debugging:** Runtime error analysis, exception handling, Streamlit troubleshooting, library integration, logic corrections.
- **Documentation:** Project proposal, problem statement, solution architecture, requirements, planning, technical reports, testing docs.
- **Knowledge Assistance:** Clarifying Python, Pandas, Plotly, Streamlit, BFSI concepts.

Groq API as the AI Recommendation Engine

Overview: Groq API is the only AI service integrated directly into the application. Unlike ChatGPT, which assisted development, Groq API operates in production.

Workflow:

1. User uploads expense data
2. Python + Pandas process transactions
3. Subscription detection identifies recurring payments
4. Subscription summary prepared
5. Data sent to Groq API
6. AI generates recommendations
7. Results displayed in Streamlit dashboard

Benefits:

- Personalized financial suggestions
- Improved decision-making
- Increased financial awareness
- Intelligent subscription optimization

Important Note: The project does **not** train its own AI model. Recommendations are generated via Groq API inference.

AI Integration Across SDLC

SDLC Phase	AI Contribution
Project Planning	Idea generation, project planning
Requirement Analysis	Scope definition, refinement
System Design	Architecture recommendations, workflow planning
Development	Programming assistance, code generation
Testing	Debugging, error analysis
Documentation	Technical writing, report generation
Deployment Preparation	Code review, implementation guidance

Benefits of AI Usage

- Faster development
- Improved code quality
- Better documentation
- Efficient debugging
- Enhanced user experience through Groq API recommendations

Limitations of AI Usage

- AI-generated code required manual review
- Suggestions needed testing before adoption
- Documentation required verification
- Recommendations are guidance, not financial advice
- Final decisions remained developer-controlled

Responsible Use of AI

- All outputs reviewed before implementation
- Code tested and validated
- Documentation aligned with actual functionality
- AI used as assistance, not replacement
- No sensitive data used for training

AI Contribution to the Project

- Faster development
- Better design decisions
- Higher-quality documentation
- Efficient debugging
- Improved technical understanding
- Intelligent financial recommendations
- Enhanced overall project quality

Comparison of AI Tools Used

Feature	ChatGPT	Groq API
Primary Purpose	Development Assistant	AI Recommendation Engine
Used During Development	Yes	No
Integrated into Application	No	Yes
Programming Assistance	Yes	No
Documentation Support	Yes	No
Generates User Recommendations	No	Yes
Final Software Component	No	Yes

Conclusion

Artificial Intelligence played a significant role in the Subscription Waste Detector. **ChatGPT** enhanced productivity during development, while **Groq API** provided AI-powered recommendations in the final application.